

robotics and embodied artificial intelligence [16, 38, 79, 132–135, 137–140]. These efforts focus on enhancing autonomous agents' capabilities in planning, reasoning, and collaboration within embodied environments. For instance, [138] proposes the Planner-Actor-Reporter paradigm for embodied reasoning and task planning. DECKARD [139] introduces the Planner-Actor-Reporter paradigm, which facilitates embodied reasoning and task planning by decoupling the agent's planning, execution, and reporting processes. TaPA [137] constructs a multi-modal dataset comprising multi-view RGB images of indoor scenes, human instructions, and corresponding plans to fine-tune LLMs. The fine-tuned models align visual perception with task planning, enabling them to generate more executable plans and significantly improving their performance in visually grounded tasks.

To overcome the physical constraints, the agents can generate executable plans and accomplish long-term tasks by leveraging multiple skills. In terms of control policies, SayCan [79] focuses on investigating a wide range of manipulation and navigation skills utilizing a mobile manipulator robot. Taking inspiration from typical tasks encountered in a kitchen environment, it presents a comprehensive set of 551 skills that cover seven skill families and 17 objects. These skills encompass various actions such as picking, placing, grasping, and manipulating objects, among others. TidyBot [136] is an embodied agent designed to personalize household cleanup tasks. It can learn users' preferences on object placement and manipulation methods through textual examples.

To promote the application of LLM-based autonomous agents, researchers have also introduced many open-source libraries, based on which the developers can quickly implement and evaluate agents

according to their customized requirements [19, 82, 127, 142–155]. For example, LangChain [147] is an open-source framework that automates coding, testing, debugging, and documentation generation tasks. By integrating language models with data sources and facilitating interaction with the environment, LangChain enables efficient and cost-effective software development through natural language communication and collaboration among multiple agent roles. Based on LangChain, XLang [145] provides a comprehensive set of tools and a fully integrated user interface. It focuses on executable language grounding, enabling the conversion of natural language instructions into code or action sequences that interact seamlessly with various environments, including databases, web applications, and physical robots. AutoGPT [82] is an agent that is fully automated. It sets one or more goals, breaks them down into corresponding tasks, and cycles through the tasks until the goal is achieved. WorkGPT [148] is an agent framework similar to AutoGPT and LangChain. By providing it with an instruction and a set of APIs, it engages in back-and-forth conversations with AI until the instruction is completed. GPT-Engineer [128] and DemoGPT [127] are open-source projects that focus on automating code generation through prompts to complete development tasks. SmolModels [126] offers a family of compact language models suitable for various tasks. AGiXT [144] is a dynamic AI automation platform that efficiently manages instructions and executes complex tasks across various AI providers, integrating adaptive memory, smart features, and a versatile plugin system. AgentVerse [156] is a versatile framework that facilitates researchers in creating customized LLM-based agent simulations efficiently. GPT Researcher [150] is an experimental application that leverages LLMs to efficiently

develop research questions, trigger web crawls to gather information, summarize sources, and aggregate summaries. BMTools [151] provides a platform for community-driven tool building and sharing. It supports various types of tools, enables simultaneous task execution using multiple tools, and offers a simple interface for loading plugins via URLs, fostering easy development and contribution to the BMTools ecosystem.

Remark. Utilization of LLM-based agents in supporting above applications may also entail risks and challenges. On one hand, LLMs themselves may be susceptible to illusions and other issues, occasionally providing erroneous answers, leading to incorrect conclusions, experimental failures, or even posing risks to human safety in hazardous experiments. Therefore, during experimentation, users must possess the necessary expertise and knowledge to exercise appropriate caution. On the other hand, LLM-based agents could potentially be exploited for malicious purposes, such as the development of chemical weapons, necessitating the implementation of security measures, such as human alignment, to ensure responsible and ethical use.

In summary, in the above sections, we introduce the typical applications of LLM-based autonomous agents in three important domains. To facilitate a clearer understanding, we have summarized the relationship between previous studies and their respective applications in Table 2.

4 LLM-based Autonomous Agent Evaluation

Similar to LLMs themselves, evaluating the effectiveness of LLM-based autonomous agents is a challenging task. This section outlines two prevalent approaches to evaluation: subjective and objective

methods. For a comprehensive overview, please refer to the right portion of Figure 5.

4.1 Subjective Evaluation

Subjective evaluation measures the agent capabilities based on human judgements [20, 22, 29, 80, 157]. It is suitable for the scenarios where there are no evaluation datasets or it is very hard to design quantitative metrics, for example, evaluating the agent’s intelligence or user-friendliness. In the following, we present two commonly used strategies for subjective evaluation.

Human Annotation: This evaluation method involves human evaluators directly scoring or ranking the outputs generated by various agents [22, 29, 104]. For example, in [20], the authors engage numerous annotators by asking 25 questions that explore their abilities across five key areas directly related to agent capabilities. In [80], annotators are asked to determine whether the specifically designed models can significantly enhance the development of rules within online communities.

Turing Test: This evaluation strategy necessitates that human evaluators differentiate between outputs produced by agents and those created by humans. If, in a given task, the evaluators cannot separate the agent and human results, it demonstrates that the agent can achieve human-like performance on this task. For instance, researchers in [29] conduct experiments on free-form Partisan text, and the human evaluators are asked to guess whether the responses are from human or LLM-based agent.

Remark. LLM-based agents are usually designed to serve humans. Thus, subjective agent evaluation plays a critical role, since it reflects human criterion. However, this strategy also faces issues such as high costs, inefficiency, and population bias. To address these issues, a growing number of researchers are in-

investigating the use of LLMs themselves as intermediaries for carrying out these subjective assessments. For example, in ChemCrow [76], researchers assess the experimental results using GPT. They consider both the completion of tasks and the accuracy of the underlying processes. Similarly, ChatEval [158] introduces a novel approach by employing multiple agents to critique and assess the results generated by various candidate models in a structured debate format. This innovative use of LLMs for evaluation purposes holds promise for enhancing both the credibility and applicability of subjective assessments in the future. As LLM technology continues to evolve, it is anticipated that these methods will become increasingly reliable and find broader applications, thereby overcoming the current limitations of direct human evaluation.

4.2 Objective Evaluation

Objective evaluation refers to assessing the capabilities of LLM-based autonomous agents using quantitative metrics that can be computed, compared and tracked over time. In contrast to subjective evaluation, objective metrics aim to provide concrete, measurable insights into the agent performance. For conducting objective evaluation, there are three important aspects, that is, the evaluation metrics, protocols and benchmarks. In the following, we introduce these aspects more in detail.

Metrics: In order to objectively evaluate the effectiveness of the agents, designing proper metrics is significant, which may influence the evaluation accuracy and comprehensiveness. Ideal evaluation metrics should precisely reflect the quality of the agents, and align with the human feelings when using them in real-world scenarios. In existing work, we can conclude the following representative evaluation metrics. (1) *Task success*

metrics: These metrics measure how well an agent can complete tasks and achieve goals. Common metrics include success rate [12, 22, 58, 60], reward/score [22, 60, 161], coverage [16], and accuracy/error rate [18, 40, 80, 101]. Depending on the scenario, accuracy may reflect aspects such as program executability [18] or task validity [101]. Higher values across these task success metrics indicate greater task completion ability. (2) *Human similarity metrics:* These metrics quantify the degree to which the agent behaviors closely resembles those of humans by emphasizing various aspects related to human traits, such as coherent [104], fluent [104], dialogue similarities with human [80] and human acceptance rate [101]. Higher similarity suggests better human simulation performance. (3) *Efficiency metrics:* In contrast to the aforementioned metrics used to evaluate the agent effectiveness, these metrics aim to assess the efficiency of agent. Commonly considered metrics encompass the cost associated with development [18] and training efficiency [16, 38].

Protocols: In addition to the evaluation metrics, another important aspect for objective evaluation is how to leverage these metrics. In the previous work, we can identify the following commonly used evaluation protocols: (1) *Real-world simulation:* In this method, the agents are evaluated within immersive environments like games and interactive simulators. The agents are required to perform tasks autonomously, and then metrics like task success rate and human similarity are leveraged to evaluate the capability of the agents based on their trajectories and completed objectives [12, 16, 22, 33, 38, 60, 86, 161, 164, 168]. By simulating real-world scenarios, this approach aims to provide a comprehensive evaluation of the agents' practical capabilities. (2) *Social evaluation:* This method utilizes

Table 3 For subjective evaluation, we use ① and ② to represent human annotation and the Turing test, respectively. For objective evaluation, we use ①, ②, ③, and ④ to represent real-world simulation, social evaluation, multi-task evaluation, and software testing, respectively. “✓” indicates that the evaluations are based on benchmarks.

Model	Subjective	Objective	Benchmark	Time
WebShop [86]	-	① ③	✓	07/2022
Social Simulacra [80]	①	②	-	08/2022
TE [101]	-	②	-	08/2022
LIBRO [159]	-	④	-	09/2022
ReAct [60]	-	①	✓	10/2022
Argyle et al. [29]	②	② ③	-	02/2023
DEPS [33]	-	①	✓	02/2023
Jalil et al. [160]	-	④	-	02/2023
Reflexion [12]	-	① ③	-	03/2023
IGLU [161]	-	①	✓	04/2023
Generative Agents [20]	①	-	-	04/2023
ToolBench [151]	-	③	✓	04/2023
GITM [16]	-	①	✓	05/2023
Two-Failures [162]	-	③	-	05/2023
Voyager [38]	-	①	✓	05/2023
SocKET [163]	-	② ③	✓	05/2023
MobileEnv [164]	-	① ③	✓	05/2023
Clembench [165]	-	① ③	✓	05/2023
Dialop [166]	-	③	✓	06/2023
Feldt et al. [167]	-	④	-	06/2023
CO-LLM [22]	①	①	-	07/2023
Tachikuma [168]	①	① ③	✓	07/2023
RocoBench [92]	-	① ③	✓	07/2023
AgentSims [34]	-	②	-	08/2023
AgentBench [169]	-	③	✓	08/2023
BOLAA [170]	-	③	✓	08/2023
Gentopia [171]	-	③	✓	08/2023
EmotionBench [172]	①	-	✓	08/2023
PTB [125]	-	④	-	08/2023

metrics to assess social intelligence based on the agent interactions in simulated societies. Various approaches have been adopted, such as collaborative tasks to evaluate teamwork skills, debates to analyze argumentative reasoning, and human studies to measure social aptitude [34, 80, 101, 163]. These approaches analyze qualities such as coherence, theory of mind, and social IQ to assess agents’ capabilities in areas including cooperation, communication, empathy, and mimicking human social behavior. By subjecting agents to complex interactive settings, social evaluation provides valuable insights into agents’ higher-level social cognition.

(3) *Multi-task evaluation*: In this method, people use a set of diverse tasks from different domains to evaluate the agent, which can effectively measure the agent generalization capability in open-domain environments [12, 29, 86, 151, 162–164, 169, 170]. (4) *Software testing*: In this method, researchers evaluate the agents by letting them conduct tasks such as software testing tasks, such as generating test cases, reproducing bugs, debugging code, and interacting with developers and external tools [159, 160, 167]. Then, one can use metrics like test coverage and bug detection rate to measure the effectiveness of LLM-based agents.

Benchmarks: Given the metrics and protocols, a crucial aspect of evaluation is the selection of appropriate benchmarks. Over time, various benchmarks have been introduced to assess the capabilities of LLM-based agents across diverse domains and scenarios. Many studies employ environments such as ALFWorld [60], IGLU [161], and Minecraft [16, 33, 38] to evaluate agent capabilities in interactive and task-oriented simulations. Tachikuma [168] evaluates LLMs' ability to infer and understand complex interactions involving multiple characters and novel objects through TRPG game logs. AgentBench [169] provides a comprehensive framework for evaluating LLMs as autonomous agents across diverse environments. It represents the first systematic assessment of LLMs as agents on real-world challenges across diverse domains. SocKET [163] is a comprehensive benchmark for evaluating the social capabilities of LLMs across 58 tasks covering five categories of social information such as humor and sarcasm, emotions and feelings, credibility, etc. AgentSims [34] is a versatile framework for evaluating LLM-based agents, where one can flexibly design the agent planning, memory and action strategies, and measure the effectiveness of different agent modules in interactive environments. ToolBench [151] focuses on assessing and enhancing language models' ability to use tools, featuring 16,464 real-world RESTful APIs and diverse instructions tailored for single- and multi-tool scenarios. WebShop [86] develops a benchmark for evaluating LLM-based agents in terms of their capabilities for product search and retrieval, which is constructed using a collection of 1.18 million real-world items. Mobile-Env [164] serves as an extendable interactive platform designed to evaluate the multi-step interaction capabilities of LLM-based agents. WebArena [173] offers a

comprehensive website environment that spans multiple domains. Its purpose is to evaluate agents in an end-to-end fashion and determine the accuracy of their completed tasks. GentBench [171] is crafted to evaluate the agent capabilities, including their reasoning, safety, and efficiency, when utilizing tools to complete complex tasks. RocoBench [92] comprises six tasks that evaluate multi-agent collaboration across diverse scenarios, emphasizing communication and coordination strategies to assess adaptability and generalization in cooperative robotics. EmotionBench [172] evaluates the emotion appraisal ability of LLMs, i.e., how their feelings change when presented with specific situations. It collects over 400 situations that elicit eight negative emotions and measures the emotional states of LLMs and human subjects using self-report scales. PEB [125] is tailored for assessing LLM-based agents in penetration testing scenarios, comprising 13 diverse targets from leading platforms. It offers a structured evaluation across varying difficulty levels, reflecting real-world challenges for agents. ClemBench [165] contains five Dialogue Games to assess LLMs' ability as a player. E2E [174] serves as an end-to-end benchmark for testing the accuracy and usefulness of chatbots.

Remark. Objective evaluation facilitates the quantitative analysis of capabilities in LLM-based agents through a variety of metrics. While current techniques can not perfectly measure all types of agent capabilities, objective evaluation provides essential insights that complement subjective assessment. Continued advancements in benchmarks and methodologies for objective evaluation will enhance the development and understanding of LLM-based autonomous agents further.

In the above sections, we introduce both subjective and objective strategies for LLM-based au-